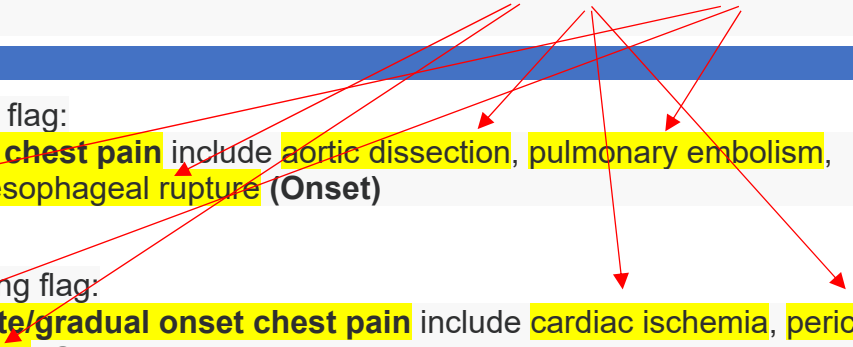

Game Times
PAIN ST
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B

1.1 - Chest Pain DDx

1) Lance hitting chest:
Chest pain can be caused by many musculoskeletal, GI, cardiac, or pulmonary conditions

2) Acutely tapering flag:
Causes of **ACUTE chest pain** include aortic dissection, pulmonary embolism, pneumothorax, & esophageal rupture **(Onset)**

3) Gradually tapering flag:
Causes of **subacute/gradual onset chest pain** include cardiac ischemia, pericarditis, pneumonia, & GERD **(Onset)**



each of these are discussed below

Cardiac Causes of Chest Pain

4) Heavy angina anvil:
Chronic stable angina & ACS (i.e., unstable angina, NSTEMI, STEMI) cause **anginal chest pain** (i.e., chest pressure, heaviness, tightness, fullness, crushing, squeezing) **(Quality)**

36) Old blacksmith:
Patients over age 55 are at increased risk for angina and ACS **(Demographics)**

I think the below (in the box) applies to all of the diseases listed above

23) Diffuse ash spots on blacksmith:
Ischemic chest pain is often **diffuse and difficult to localize (Location)**

24) Blacksmith's covered left arm/shoulder:
Ischemic chest pain classically **radiates to the left shoulder, arm, and/or jaw (Location)**

45) Blacksmith blowing puff of air:
Ischemic chest pain can manifest as **dyspnea**, especially in women, elderly, or diabetic patients **(ROS)**

52) Red pipe filled with yellow gunk:
Patients with ischemic chest pain often have risk factors for **atherosclerosis** (e.g., DM, HLD, HTN, smoking) **(PMHx)**

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28) Exertion at the anvil:
Chronic **stable angina** **is provoked by exertion** and relieved with rest **(Provocation/Palliation)**

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37) Young lady with medal:
Vasospastic angina is more common in **women and people younger than age 50 (Demographics)**

5) Anvil medal:
Vasospastic angina (aka **Prinzmetal angina**) also causes anginal chest pain **(Quality)**

82) “Pain ST” sign:
ACS, vasospastic angina, & stress-induced cardiomyopathy are often associated with **ST changes on ECG (ECG)**

82) “Pain ST” sign:
ACS, vasospastic angina, & stress-induced cardiomyopathy are often associated with **ST changes on ECG (ECG)**

29) Sleeping anvil medal wearer:
Vasospastic angina occurs at rest, classically **at night during sleep (Provocation/Palliation)**

65) Hot cocoa:
Cocaine use can precipitate ACS or vasospastic angina **(Social Hx)**

35) Episodic events calendar:
Vasospastic angina & diffuse esophageal spasm cause intermittent pain with **no obvious pattern (Provocation/Palliation)**

66) BP cuff:
Hypotension can occur in life-threatening causes of chest pain (e.g., ACS, aortic dissection, PE, tension PTX, cardiac tamponade, esophageal rupture) **(Vital Signs)**

68) Distended blue jugs:
Conditions that cause right-sided heart failure (e.g., **ACS**, PE) can produce **JVD (Physical Exam)**

65) Hot cocoa:
Cocaine use can precipitate ACS or vasospastic angina **(Social Hx)**

71) Murmur lines from anvil hammer:
ACS can present with new-onset **murmur (Cardiac Auscultation)**

73) **S**lushi**3** cup:
ACS may present with **S3 gallop (Cardiac Auscultation)**

77) Crackly velcro on armor on anvil:
ACS may present with **crackles** 2/2 acute left-sided heart failure **(Lung Auscultation)**

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86) T-bone steak:
Acute MI & stress-induced cardiomyopathy can cause **elevated troponin levels (Labs)**

6) Stressed lady next to anvil:

Stress-induced (aka Takotsubo) cardiomyopathy also causes anginal chest pain **(Quality)**

30) Stressed-out lady:
Stress-induced cardiomyopathy is typically **precipitated by intense emotional stress (Provocation/Palliation)**

38) Old stressed-out woman:
Stressed-induced cardiomyopathy primarily occurs in **postmenopausal women (Demographics)**

82) “Pain ST” sign:
ACS, vasospastic angina, & stress-induced cardiomyopathy are often associated with **ST changes on ECG (ECG)**

86) T-bone steak:
Acute MI & stress-induced cardiomyopathy can cause **elevated troponin levels (Labs)**

7) Hand grabbing heart:
Acute cardiac tamponade can present as sudden onset chest pain and dyspnea **(Quality)**

62) Arm cast on knight with crushed heart sigil:
Blunt or penetrating **trauma to the chest** is a common cause of acute cardiac tamponade **(HPI)**

63) Dissecting lance hitting crushed heart sigil:
Type A **aortic dissection** can cause acute cardiac tamponade **(HPI)**

64) Invasive cape drawstring:
Invasive procedures (e.g., cardiac cath) can cause acute cardiac tamponade **(HPI)**

66) BP cuff:
Hypotension can occur in life-threatening causes of chest pain (e.g., ACS, aortic dissection, PE, tension PTX, cardiac tamponade, esophageal rupture) **(Vital Signs)**

70) Triangular B shield:
Cardiac tamponade can produce **Beck's Triad** (i.e., hypotension, elevated JVP, muffled heart sounds) **(Physical Exam)**

84) Knocked over ECG flag:
Cardiac tamponade can present with **diffuse low voltages** and electrical alternans on ECG **(ECG)**

85) Alternating high-low fence posts:
Cardiac tamponade can present with diffuse low voltages and **electrical alternans** on ECG **(ECG)**

96) Ultra-sounding horn next to crushed heart knight:
Bedside **ultrasound** can help diagnose cardiac tamponade & tension PTX in patients with chest pain **(Imaging)**

8) Dissected lance with torn chest:
Aortic dissection causes acute onset of “ripping” or “tearing” chest pain **(Quality)**

26) Tearing back cape:
Aortic dissection can present with tearing chest pain that **radiates to the back** **(Location)**

53) Steam coming out of ears:
HTN is the most important risk factor for aortic dissection **(PMHx)**

54) Martian helmet:
Inherited collagen diseases (e.g., **Marfan syndrome**, Ehlers-Danlos syndrome) are risk factors for aortic dissection **(PMHx)**

63) Dissecting lance hitting crushed heart sigil:
Type A **aortic dissection** can cause acute cardiac tamponade **(HPI)**

66) BP cuff:
Hypotension can occur in life-threatening causes of chest pain (e.g., ACS, aortic dissection, PE, tension PTX, cardiac tamponade, esophageal rupture) **(Vital Signs)**

72) Murmur sound from dissection horse:
Aortic dissection can present with new-onset **murmur (Cardiac Auscultation)**

95) Skull & X-bones shield on knight with dissected lance:
CXR can help diagnose aortic dissection **(Imaging)**

97) CT cat next to knight with dissected lance:
CT can help diagnose aortic dissection **(Imaging)**

Pulmonary Causes of Chest Pain

9) Inspiring with shark-tooth necklace:
Many conditions (e.g., PE, pneumothorax, pneumonia, pleuritis, pericarditis) can cause **pleuritic chest pain** (i.e., sharp, worsened by inspiration) **(Quality)**

10) Shark-tooth necklace with lung tree:
Many conditions (e.g., **PE**, pneumothorax, pneumonia, pleuritis, pericarditis) can cause pleuritic chest pain (i.e., sharp, worsened by inspiration) **(Quality)**

46) Guy with collapsed sail chestplate blowing puff of air:
Pulmonary causes of chest pain (e.g., **PE**, pneumothorax, pleuritis, pneumonia) also commonly cause **dyspnea (ROS)**

49) Guy with dirty lung stains blowing puff of air:
Pulmonary causes of chest pain (e.g., **PE**, pneumothorax, pleuritis, pneumonia) also commonly cause **dyspnea (ROS)**

51) Red, swollen leg:
PE is often associated with **symptoms of DVT (i.e., leg pain and swelling) (ROS)**

55) Wheelchair:
Risk factors for DVT/PE include **prolonged immobilization**, recent surgery, & hypercoagulability (e.g., malignancy) **(PMHx)**

56) Surgery scalpel:
Risk factors for DVT/PE include prolonged immobilization, **recent surgery**, & hypercoagulability (e.g., malignancy) **(PMHx)**

57) Cancer crab:
Risk factors for DVT/PE include prolonged immobilization, recent surgery, & hypercoagulability (e.g., **malignancy) (PMHx)**

66) BP cuff:
Hypotension can occur in life-threatening causes of chest pain (e.g., ACS, aortic dissection, PE, tension PTX, cardiac tamponade, esophageal rupture) **(Vital Signs)**

69) Man in wheelchair holding distended blue jug:
Conditions that cause right-sided heart failure (e.g., ACS, **PE**) can produce **JVD (Physical Exam)**

87) CT cat on top of well:
Perform **CT** angiography in patients with high pre-test probability of PE **(Imaging)**

88) Dual D-shaped twigs below well:
Obtain **D-dimer** in patients with low pre-test probability of PE **(Labs)**

11) Shark-tooth necklace with collapsed lung sail:
Many conditions (e.g., PE, **pneumothorax**, pneumonia, pleuritis, pericarditis) can cause pleuritic chest pain (i.e., sharp, worsened by inspiration) **(Quality)**

47) Guy blowing puff of air at shield with lung tree:
Pulmonary causes of chest pain (e.g., PE, **pneumothorax**, pleuritis, pneumonia) also commonly cause **dyspnea (ROS)**

58) Tall and thin man with collapsed sail chestplate:
Primary spontaneous PTX classically occurs in **tall, thin, previously healthy young males (PMHx)**

59) Tobacco pipe:
Smoking is a risk factor for primary spontaneous PTX **(Social Hx)**

60) Blue bloated helmet:
COPD is a risk factor for secondary spontaneous PTX **(PMHx)**

66) BP cuff:
Hypotension can occur in life-threatening causes of chest pain (e.g., ACS, aortic dissection, PE, tension PTX, cardiac tamponade, esophageal rupture) **(Vital Signs)**

75) No sound on collapsed sail side:
PTX can produce **absent breath sounds on the affected side (Lung Auscultation)**

78) Resonant drum:
PTX produces **increased resonance to percussion** on the affected side **(Physical Exam)**

79) Bubble wrap cape:
PTX can cause **subcutaneous emphysema (Physical Exam)**

93) Skull & X-bones shield next to collapsed sail armor:
CXR can help diagnose PTX **(Imaging)**

96) Ultra-sounding horn next to crushed heart knight:
Bedside **ultrasound** can help diagnose cardiac tamponade & tension PTX in patients with chest pain **(Imaging)**

12) Shark-tooth necklace with dirty lung stains:

Many conditions (e.g., PE, pneumothorax, **pneumonia**, pleuritis, pericarditis) can cause pleuritic chest pain (i.e., sharp, worsened by inspiration) **(Quality)**

42) Flame bandana on guy with dirty lung stains:
Pneumonia commonly presents with **fever (ROS)**

76) Crackly velcro above guy with dirty lung stains:
Pneumonia commonly present with **crackles (Lung Auscultation)**

89) White birds next to dirty lung stains:
Pneumonia can produce an **elevated WBC count (Labs)**

92) Skull & X-bones shield held by guy with dirty lung stains:
CXR can help diagnose PNA **(Imaging)**

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13) Shark-tooth necklace with lupus wolf:

Many conditions (e.g., PE, pneumothorax, pneumonia, **pleuritis**, pericarditis) can cause pleuritic chest pain (i.e., sharp, worsened by inspiration) **(Quality)**

48) Wolf attack victim blowing puff of air:
Pulmonary causes of chest pain (e.g., PE, pneumothorax, **pleuritis**, pneumonia) also commonly cause **dyspnea (ROS)**

14) Shark-tooth necklace with inflamed sac:
Many conditions (e.g., PE, pneumothorax, pneumonia, pleuritis, **pericarditis**) can cause pleuritic chest pain (i.e., sharp, worsened by inspiration) **(Quality)**

31) Leaning over inflamed case:
Pericarditis pain is classically worse when lying supine & **relieved by sitting up and leaning forward** **(Provocation/Palliation)**

44) Flame bandana on violinist with inflamed case:
Pericarditis commonly presents with **fever (ROS)**

74) Bow rubbing on violin strings:
Pericarditis may present with a **pericardial friction rub (Cardiac Auscultation)**

83) Multiple “ST” signs:
Pericarditis may present with **diffuse ST elevations on ECG (ECG)**

90) White birds next to inflamed case:
Pericarditis can produce an **elevated WBC count (Labs)**

Gastrointestinal Causes of Chest Pain

32) Food pattern on esophageal banner:
GI causes of chest pain may be provoked OR palliated by **food (Provocation/Palliation)**

50) Princess choking doll:
Esophageal causes of chest pain (e.g., GERD, esophageal motility disorders, esophagitis, hiatal hernia, rupture) can cause **dysphagia or odynophagia (ROS)**

15) Refluxing goblet:

GERD classically presents with burning retrosternal chest pain **(Quality)**

16) Flaming chest:
GERD classically presents with **burning retrosternal chest pain** **(Quality)**

17) Anvil next to refluxing goblet:
GERD can also present with squeezing retrosternal chest pain similar to **angina** **(Quality)**

25) Flame pattern on queen's dress:
Chest pain due to GERD can also **radiate to the back, neck, jaw, or arms** **(Location)**

33) King sleeping next to reflux goblet:
GERD symptoms are typically **worse at night during sleep** **(Provocation/Palliation)**

34) Prince popping pills:
GERD symptoms are typically **alleviated with antacids** **(Provocation/Palliation)**

50) Princess choking doll:
Esophageal causes of chest pain (e.g., GERD, esophageal motility disorders, esophagitis, hiatal hernia, rupture) can cause **dysphagia or odynophagia (ROS)**

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50) Princess choking doll:
Esophageal causes of chest pain (e.g., GERD, esophageal motility disorders, esophagitis, hiatal hernia, rupture) can cause **dysphagia or odynophagia (ROS)**

18) Princess struggling to climb down banner:
Other esophageal disorders (e.g., **esophageal motility disorders**, esophagitis, hiatal hernia) can present with GERD-like heartburn **(Quality)**

50) Princess choking doll:
Esophageal causes of chest pain (e.g., GERD, esophageal motility disorders, esophagitis, hiatal hernia, rupture) can cause **dysphagia or odynophagia (ROS)**

50) Princess choking doll:
Esophageal causes of chest pain (e.g., GERD, esophageal motility disorders, esophagitis, hiatal hernia, rupture) can cause **dysphagia or odynophagia (ROS)**

19) Ulcerated banner:
Other esophageal disorders (e.g., esophageal motility disorders, **esophagitis**, hiatal hernia) can present with GERD-like heartburn **(Quality)**

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50) Princess choking doll:
Esophageal causes of chest pain (e.g., GERD, esophageal motility disorders, esophagitis, hiatal hernia, rupture) can cause **dysphagia or odynophagia (ROS)**

20) Bulging banner:
Other esophageal disorders (e.g., esophageal motility disorders, esophagitis, **hiatal hernia**) can present with GERD-like heartburn **(Quality)**

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21) Twisting esophageal hat:
Diffuse esophageal spasm (aka DES) can present with GERD-like heartburn or angina **(Quality)**

35) Episodic events calendar:
Vasospastic angina & diffuse esophageal spasm cause intermittent pain with **no obvious pattern** **(Provocation/Palliation**

50) Princess choking doll:
Esophageal causes of chest pain (e.g., GERD, esophageal motility disorders, esophagitis, hiatal hernia, rupture) can cause **dysphagia or odynophagia (ROS)**

22) Esophageal horn bursting:
Esophageal rupture (e.g., Boerhaave syndrome) presents with sudden, severe retrosternal chest pain **(Quality)**

66) BP cuff:
Hypotension can occur in life-threatening causes of chest pain (e.g., ACS, aortic dissection, PE, tension PTX, cardiac tamponade, esophageal rupture) **(Vital Signs)**

43) Flame bandana on musician with bursting horn:
Esophageal perforation often causes mediastinitis and **fever (ROS)**

67) Septic sewer lid:
Esophageal perforation can cause **sepsis** (e.g., fever, hypotension, tachycardia, tachypnea, cyanosis) **(Vitals)**

80) Bubble wrap scarf:
Esophageal perforation can cause **subcutaneous emphysema (Physical Exam)**

91) White birds above ruptured horn:
Esophageal rupture can cause mediastinitis, which produces an **elevated WBC count (Labs)**

94) Skull & X-bones shield held by kid with ruptured horn:
CXR can help diagnose esophageal rupture **(Imaging)**

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81) Up arrow on intestinal organ vendor:
Perform a full abdominal exam as abdominal disease can cause **referred chest pain (Physical Exam)**

Musculoskeletal Causes of Chest Pain

27) Pinpoint strike on skeletal target:
Musculoskeletal pain (e.g., rib fracture, costochondritis) is **well-localized (i.e., patients can point to one spot) (Location)**

Autoimmune Causes of Chest Pain

39) Rashy guy being attacked by wolf:
Autoimmune causes of chest pain (e.g., lupus pleuritis) are associated with other systemic symptoms (e.g., **rash**, arthralgias, fever) **(ROS)**

40) Wrapped joints:
Autoimmune causes of chest pain (e.g., lupus pleuritis) are associated with other systemic symptoms (e.g., rash, **arthralgias**, fever) **(ROS)**

41) Flame bandana:
Autoimmune causes of chest pain (e.g., lupus pleuritis) are associated with other systemic symptoms (e.g., rash, arthralgias, **fever) (ROS)**